

8

Protective Relays

8.1 Introduction:

- An electrical power system consists of generators, Transformers, transmission and distribution line. The system equipments are likely to damage during fault or abnormal condition in any part of the power system, if proper protection system is not provided.
- If a fault occurs in any part of the power system, an automatic protective device is needed to isolate the faulty element as quickly as possible.
- A protective system includes, circuit breakers and protective relays to isolate the faulty section of the system from healthy sections.
- The function of a protective relay is to detect an abnormal condition in the power system and issue trip signal to the circuit breaker to disconnect the faulty element
- The basic electrical quantities likely to change during abnormal condition are voltage, current, frequency and phase angle (Direction)
- The protective relays utilize one or more of these quantities to detect abnormal conditions in a power system.

Back-up Relay: a back up relay operates after a slight delay, if the main relay fails to operate.

Back up protection: The back up protection is designed to clear the fault if the primary protection fails. It acts as a second line of defense.

Primary protection: If a fault occurs, it is duty of the primary protective scheme to clear the fault and it acts as a first line of defense. If it fails, the back – up protection clears the fault.

Stability: This is the ability of the protective system to remain inoperative under all load conditions, and also in case of external faults. The relay should remain stable when a heavy current due to an external fault is flowing through it.

Pick up level: The value of actuating quantity above which relay operates.

Reset level: The value of actuating quantity below which the relay is de-energized and returns to its original position.

Relay: Relay is a sensing devices, this will sense the abnormal condition such as short circuit fault in the system and gives trip signal for circuit breaker

Operating Time: The time which elapses between the instant when the actuating quantity exceeds the pickup value to the instant when the relay contacts get closed.

Reset Time: The time which elapses between the instant when the actuating quantity becomes less than the reset value to the instant when the relay contact returns to its normal position.

Primary Relays: The relays which are connected directly in the circuit to be protected.

Secondary Relays: The relays which are connected in the circuit to be protected through current and potential transformers.

Auxiliary Relays: Relays which operate in response to the opening or closing of its operating circuit to assist another relay in the performance of its function. This relay may be instantaneous or may have a time delay.

8.2 Basic objectives of system protection

The fundamental objective of system protection is to provide isolation of a problem area in the power system quickly, so that the damage to the rest of the system is minimized and as much as possible is left intact.

Thus protection does not mean prevention, but rather, minimizing the duration of the trouble and limiting the damage, outage time, and related problems that may result otherwise.

The basic objectives of system protection are

- (i) Speed of operation
- (ii) Sensitivity
- (iii) Selectivity
- (iv) Reliability



- (v) Simplicity
- (vi) Stability
- (vii) Economics

Speed of operation

The protection system should ensure minimum fault duration, consequent minimum damage to equipment and system stability.

If the fault clearing is delayed, the healthy equipments of the system will damage due to excessive thermal and mechanical stress. So the protection system should isolate faulty section from healthy system as quickly as possible.

Sensitivity

The protection system should be able to detect minimum possible value of fault current and issue trip signal to circuit breaker.

Selectivity (Discrimination)

The protection system should ensure maximum continuity of service and minimum system interruption

The relay should be able to discriminate the fault in its own zone of protection or outside the zone of protection.

Reliability

It is the assurance that the protection system will perform 100% correctly

It has two aspects

1. Dependability
2. Security

Dependability: It should operate when it is required

Security: It should not operate when it is not required

Simplicity

The protection system should have minimum number of equipment and associated circuitry to achieve the protection objectives

Stability

Relay operation must be in stable mode even it is carrying the fault current

The relay should be able to carry the fault current without damaging itself till the point of interruption.

Economics

Protection should provide maximum protection at minimal total cost.

8.3 Classification of relays

• Classification based on technology

Electromechanical Relays: These relays were the earliest forms of relay used for protection of power system. They work on principle of mechanical force causing an operation of relay.

Static Relays: These relays don't have moving parts. The design of static relays is based on the use of Analog electronics instead of coils and magnets to create relay characteristics.

Digital Relays: To implement the relay functions, Analog circuits used in static relays are replaced with microprocessors and micro controllers in digital relays

Numerical Relays: These relays can be viewed as natural developments of digital relays as a result of advances in technology. These relays use DSP processor as a computational hardware, together with the associated software tools.

• Classification based on application

1. over current relays
2. Directional relays
3. Distance relays
4. Differential relays

8.4. Operating principle of Electro mechanical Relays

Basically there are two different operating principles of relays

- (i) electro magnetic attraction
- (ii) electro magnetic induction

8.4.1 Electromagnetic attraction relay

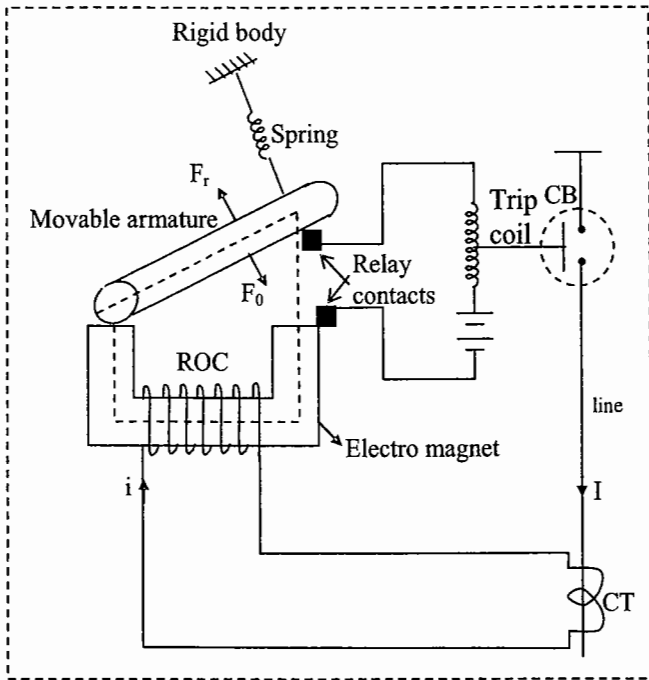


Fig. 8.1: Attracted armature Relay

- A restraining force has to be produced exactly in opposite direction of operating force to restrict relay operation under normal condition.
- Field energy stored in electro magnetic circuit is function of air gap length (x)

$$W = \frac{1}{2} Li^2$$

$$L = \frac{\mu N^2 a}{l}$$

l = length of magnetic path

N = No. of turns of coil

a = area of cross section of coil

- The inductance 'L' changes with respect to contact gap

$$\therefore W(x) = \frac{1}{2} L(x) i^2$$

- Torque produced by electro magnetic circuit,

$$T_{op} = \frac{\partial W}{\partial x}$$

$$T_{op} = \frac{1}{2} i^2 \frac{dL}{dx}$$

$$i = I_m \sin \omega t, F_0 = \frac{1}{2} (I_m \sin \omega t)^2 \frac{dL}{dx}$$

$$T_{op} = \frac{1}{2} \frac{dL}{dx} I_m^2 \sin^2 \omega t$$

$$= \frac{1}{2} \frac{dL}{dx} I_m^2 \left(\frac{1 - \cos 2\omega t}{2} \right)$$

$$= \frac{1}{4} \frac{dL}{dx} I_m^2 (1 - \cos 2\omega t)$$

$$T_{op} (Avg) = \frac{1}{4} \frac{dL}{dx} I_m^2 \quad (\cos 2\omega t)_{avg} = 0$$

- For a constant 'x', $\frac{dL}{dx}$ is constant

$$T_{op} \propto I^2$$

$$T_{res} = K I^2$$

$$\text{Resultant torque, } T = T_{op} - T_{res} \\ = K_1 I^2 - K$$

Where K is the torque produced by spring

- This type of relay will be used for both AC and DC relaying purposes.

8.4.2. Induction type relays:

Applicable only for AC relaying purpose. There are two types of induction relays

1. Induction-disc type element
 - (i) Shaded pole structure
 - (ii) Watt hour meter/energy meter structure
2. Induction cup type element

Shaded pole induction disc type

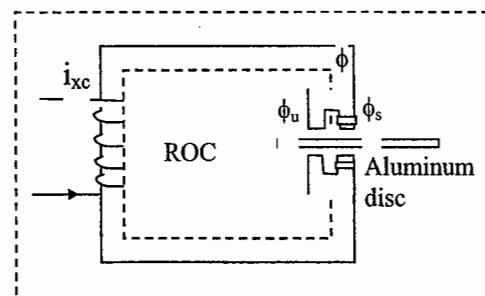


Fig. 8.2: Shaded pole Induction Relay

For the relay application, continuous rotation of disc is not required. To restrict the rotation of disk during normal operation, a restraining torque is

produced by using spring. To damp out oscillations made by disc a permanent magnet (PM) is provided.

$$T \propto \phi_s \phi_u \sin \alpha$$

$$T = I^2 \sin \alpha$$

$$T = K_1 I^2 \quad K = \text{constant}$$

Watt-hour meter (energy meter) disc type element

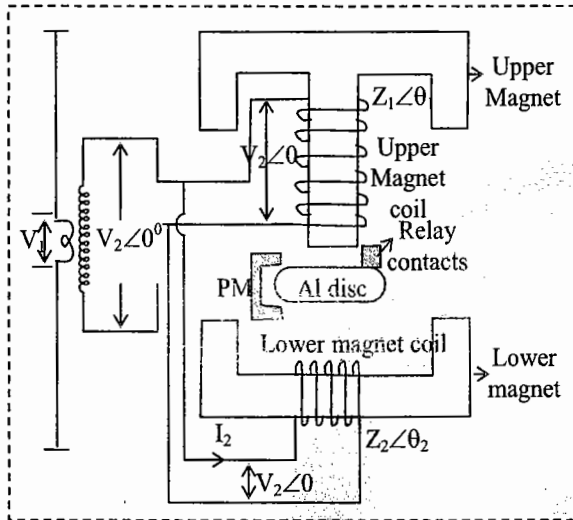


Fig. 8.3: Watt-hour meter type Induction Relay

Upper magnet coil	Lower magnet coil
$V_2 \angle 0^\circ$	$V_2 \angle 0^\circ$
$\downarrow$	$\downarrow$
$\bar{I}_1 = \frac{V_1 \angle 0}{Z_1 \angle \theta_1}$	$\bar{I}_2 = \frac{V_1 \angle 0}{Z_2 \angle \theta_2}$
$\downarrow$	$\downarrow$
$I_1 \angle -\theta_1$	$\Rightarrow I_2 \angle -\theta_2$
$\downarrow$	$\downarrow$
$\phi_1 \angle -\theta_1$	$\phi_2 \angle -\theta_2$

Torque produced on Aluminum disc
 $T \propto \phi_1 \cdot \phi_2 \sin(\theta_1 - \theta_2)$

- If $\theta_1 - \theta_2 = 0$ (or) $\theta_1 = \theta_2$, then torque produced by Aluminum disc is zero and disc will not rotate.
- To get rotation Aluminum disc $\theta_1 \neq \theta_2$ or $\frac{X}{R}$ ratios of upper coil and lower coil maintained different

Induction cup type relay:

- Operated based on induction motor principle, i.e., the torque produced is due to interaction of two rotating magnetic fields.
- Rotating body is a hollow Aluminum cup
- This is constructed either for 4 pole or 8 pole.

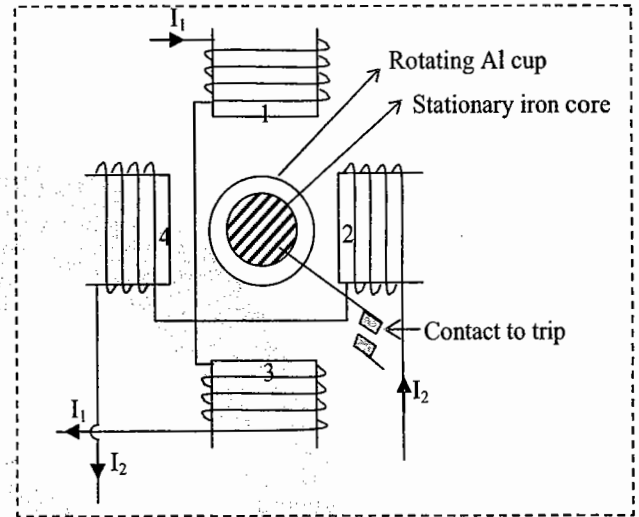


Fig. 8.3: Watt-hour meter type Induction Relay

- Size of cup is very small and it's weight and inertia is very less
- Torque/weight ratio is very high
- Typical operating time = 0.01 sec
- This is high speed relay
- Sensitivity of relay is very high and the ratio between reset level to pickup level is around 0.99.
- Cup type relays are used in high speed relaying applications

Applications of induction type elements:

- All the three elements will be used as inverse time over current relays.
- Watt-hour meter and cup type elements will be used as directional relays
- In the distance relay both operating forces and restraining forces will be produced by system quantities by reconnecting the coils.



8.4.3. Thermal relay:

Thermal relay used to protect the rotating machines from over heating. (Cause for the over heating is over load on the system)

Types of thermal relays

1. Thermo couple/Thermistor
2. RTD (Resistance temperature detector)
3. Bimetallic strip

Universal Torque Equation

$$T = K_1 I^2 + K_2 V^2 + K_3 VI \cos(\theta - \tau) + K$$

$\Downarrow$
Current
unit

$\Downarrow$
Voltage
unit

$\Downarrow$
Directional
unit

$\Downarrow$
Spring
torque

8.5. Over current Relays

From universal torque equation

$$T = K_1 I^2 + K_2 V^2 + K_3 VI \cos(\theta - \tau) + K$$

$$K_1 = +ve, K_2 = 0, K_3 = 0, K = -ve$$

$$T = K_1 I^2 - K$$

Under threshold

$$T = 0; T_{op} = T_{res}$$

$$K_1 I^2 - K = 0$$

$$K_1 I^2 = K$$

$$I = \sqrt{\frac{K}{K_1}}$$

$$I_{pick\ up} = \sqrt{\frac{K}{K_1}}$$

During Operation

$$T > 0; T_{op} > T_{res}$$

$$K_1 I^2 - K > 0$$

$$K_1 I^2 > K$$

$$I > \sqrt{\frac{K}{K_1}}$$

$$I > I_{pick\ up}$$

$\therefore$ If current flowing through the relay operating coil is greater than the pick up value of current Relay operates.

$$I > I_{pick\ up} \rightarrow \text{Relay operates}$$

$$I < I_{pick\ up} \rightarrow \text{Relay Not operates}$$

$$I = I_{pick\ up} \rightarrow \text{Relay is on Threshold.}$$

Current Setting: It is defined as the ratio of pick up current of relay to the relay current rating

$$\text{Current setting (C.S) \%} = \frac{I_{pick\ up}}{I_{relay}} \times 100$$

Note: Current settling is also expressed in Amperes

If current setting is in Amperes then,

Pickup current = current setting (A)

= Plug setting (A)

= Set value of current (A)

8.5.1 Plug Setting Multiplier (P.S.M)

$$P.S.M = \frac{\text{primary fault current of C.T}}{C.T \text{ Ratio} \times \text{Relay current setting (A)}}$$

$$= \frac{\text{secondary fault current}}{\text{Relay current setting (A)}} \quad 3$$

Relay current setting (A)

$$= \frac{C.S(\%)}{100} \times \text{Relay current rating.}$$

Note: If relay current rating is not given, the secondary current of C.T is reference to calculate current settling.

8.5.2 Characteristics of over current Relay

It is a curve between P.S.M versus time of operation of over current relay

$$T_{op} = KI^2; T_{res} = K \text{ (constant)}$$

$$T = KI^2 - K$$

- If fault current increases, P.S.M increases net torque (T) increases, speed of the disc increases and time of operation decreases.
- So There is a inverse relation between P.S.M and time of operation of relay

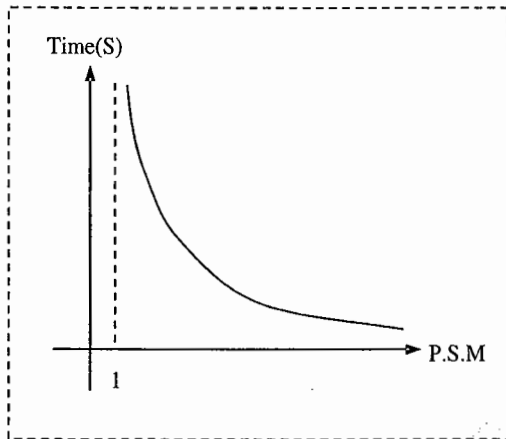
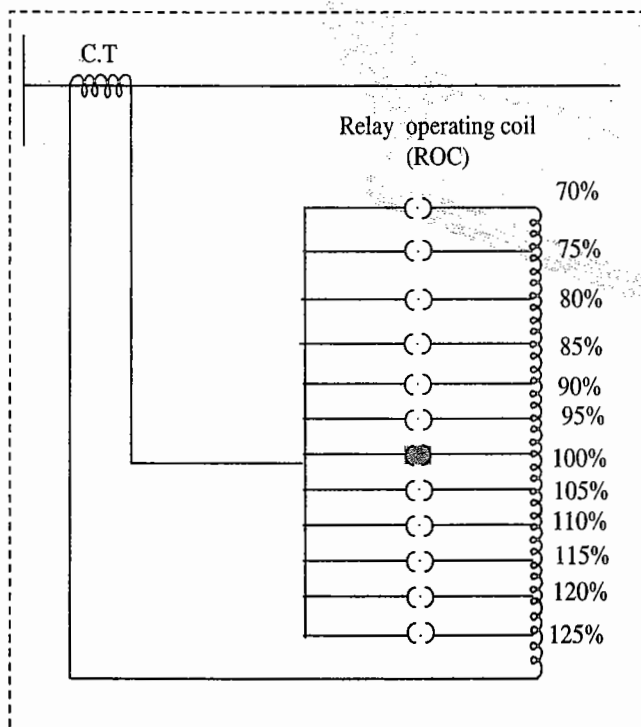


Fig. 8.5: Characteristics of Over Current Relay

8.5.3 Time Multiplier Setting (T.M.S):

- To control actual operating time of over current relay, Manufacturers provide a provision for changing the operating time.
- Operating time of relay = T.M.S $\times$ Operating time of relay with 100 % T.M.S
- To adjust current setting, the number of turns in relay coil is to be adjusted.
- By adjusting back stop position operating time of relay is adjusted.



8.5.4 Types of Over Current Relays

Depending upon the time of operation the relays are categorized as:

- Instantaneous over current relay,
- Definite time over current relay
- Inverse definite minimum time (IDMT) over current relay
- Very inverse relay,
- Extremely inverse relay.

Instantaneous over-current relay

Here there is no intentional time delay is provided for the operation of Relay. The time of operation of such relays is approximately 0.1 sec. This characteristic can be achieved with the help of hinged armature relays.

Definite time-current relay

Here the operating time is constant through out the operation. The operating time of Relay is independent of PSM

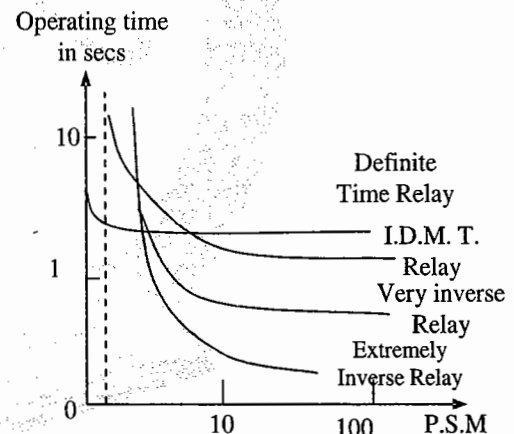


Fig. 2.6: Characteristics of Over Current Relays

Inverse Definite Minimum Time Relay

In IDMT Relay the operating time is approximately inversely proportional to the fault current near pick up value and becomes substantially constant slightly above the pick up value of the relay. This is achieved by using a core of the electromagnet which gets saturated for currents slightly greater than the pick up current.

Very inverse relay

In Very Inverse Relay the saturation of the core occurs at a later stage. The characteristics relay is as shown in Fig. 5.6. The time current characteristic is inverse over a greater range and after saturation tends to definite time.

Extremely inverse relay

In Extremely Inverse Relay the saturation of the core occurs at a later stage, the characteristic assumes the shape as shown in Fig. 5.6. The equation describing extremely inverse relay is approximately of the form $I^2 t = K$, where I is the operating current and t the operating time.

8.6 Directional Relays

From Universal torque equation

$$T = K_1 I^2 + K_2 V^2 + K_3 VI \cos(\theta - \tau) + K$$

For directional relays

$$K_1 = 0, K_2 = 0, K_3 = +ve, K = -ve$$

$$T = K_3 VI \cos(\theta - \tau) - K \dots\dots\dots (i)$$

The restraining torque ensure that for operation of directional relay a minimum current is required in the set direction

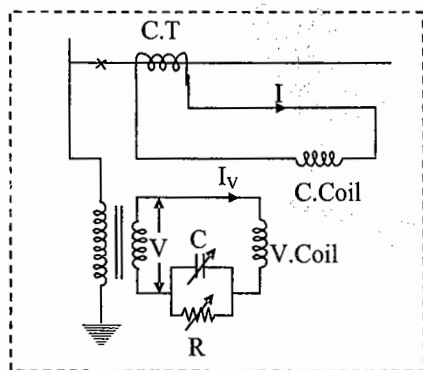


Fig. 8.7: Zero degree connection of directional relay

V – rms value of voltage fed to the voltage coil of directional relay

I – rms value of current fed to the current coil of directional relay

θ - angle between V & I

τ - maximum torque angle

- By changing the current coil connection the set direction of directional relay can be changed.

$$\text{If } (\theta - \tau) = \text{constant}$$

$$\cos(\theta - \tau) = \text{constant}$$

$$\text{equ. (i)} \Rightarrow K_3 VI - |K|$$

Under threshold

$$T = 0 ; T_{op} = T_{res}$$

$$K_3 VI - |K| = 0$$

$$VI = \frac{|K|}{K_3} \dots\dots\dots (ii)$$

Equ. (ii) gives constant product characteristics for directional relay.

During operation

$$T > 0, T_{op} > T_{res}$$

$$K_3 VI - K > 0$$

$$VI > \frac{K}{K_3}$$

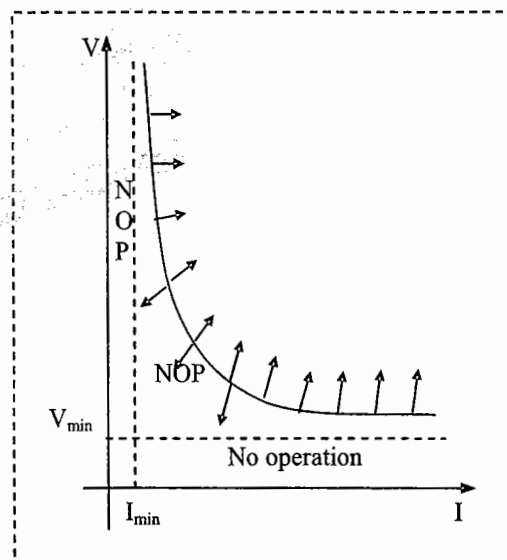


Fig. 8.8: Constant product characteristics of directional relay

Conclusion

- If $I < I_{min}$ (in set direction) and for any value of voltage the relay will not operate. (This is expected)
- If $V < V_{min}$ and for any value of current relay not operates (This should be avoided)
- If there is a fault very near to relay location, the voltage fed to the relay is less than the minimum voltage required for the relay operation and directional relay does not operate.

Dead zone

The maximum distance upto which the voltage fed to the relay is less than the minimum voltage required for relay operation is called dead zone of directional relay.

- If there is a fault any where in the dead zone the relay will not operate.
- To avoid the dead zone problem the voltage coil of directional relay should not be connected to the same phase where current coil is connected.

Phasor diagram

$$T_{op} = K_3 VI \cos(\theta - \tau)$$

or

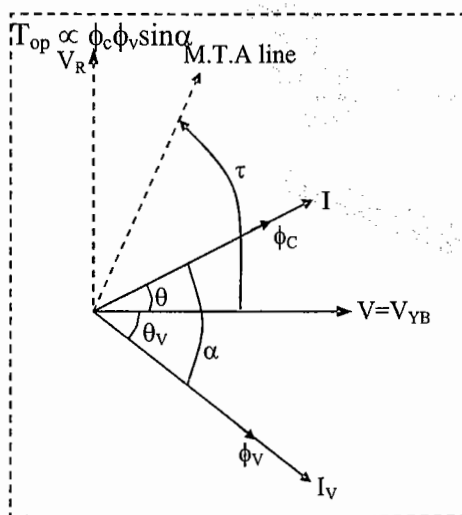


Fig. 8.9: Phasor diagram of directional relay

- The MTA line is always in quadrature with voltage coil current or voltage coil flux.

- Angle between voltage coil voltage and voltage coil current is adjusted with the help of phase shifting network. So it is possible to adjust the maximum torque angle.

- If $\theta_v = 90^\circ$, maximum torque angle, $\tau = 0^\circ$
- If $\theta_v = 0^\circ$, maximum torque angle, $\tau = 90^\circ$
- If $\theta_v = 45^\circ$, maximum torque angle $\tau = 45^\circ$

Polar characteristics

If restraining torque assumed as zero then total torque $T = T_{op}$

$$T = K_3 VI \cos(\theta - \tau)$$

Under threshold

$$T = 0 \Rightarrow K_3 VI \cos(\theta - \tau) = 0$$

$$\theta - \tau = \pm \frac{\pi}{2} \dots\dots\dots (iii)$$

Equ. (iii) gives polar characteristics of directional relay.

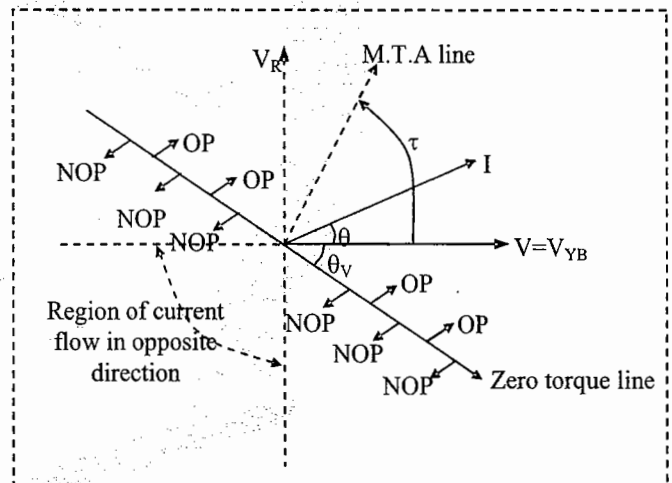


Fig. 8.10: Polar characteristics of directional relay

- Zero torque line is in phase with voltage coil current or voltage coil flux.
- Zero torque line and maximum torque lines are orthogonal to each other.

8.7 Distance Relays

- Impedance Relay
- Reactance Relay
- Mho's Relay

8.7.1 Impedance Relay

From Universal torque equation

$$T = K_1 I^2 + K_2 V^2 + K_3 VI \cos(\theta - \tau) + K$$

For impedance Relay

$$K_1 = +ve, K_2 = -ve, K_3 = 0, K = 0$$

$$T = KI^2 - K_2 V^2 \dots\dots\dots (iv)$$

- Current unit is providing operating torque
- Voltage unit is providing restraining torque
- This relay is also called voltage restrained over current relay.

Under threshold

$$T = 0, T_{op} = T_{res}$$

$$K_1 I^2 - K_2 V^2 = 0$$

$$K_1 I^2 = K_2 V^2$$

$$\frac{V^2}{I^2} = \frac{K_1}{K_2}$$

$$\frac{V}{I} = \sqrt{\frac{K_1}{K_2}}$$

$$Z_{set} = \sqrt{\frac{K_1}{K_2}} \dots\dots\dots (v)$$

When relay is operating

$$T > 0, T_{op} = T_{res}$$

$$K_1 I^2 > K_2 V^2$$

$$\frac{V^2}{I^2} < \frac{K_1}{K_2}$$

$$\frac{V}{I} < \sqrt{\frac{K_1}{K_2}}$$

$$Z_{seen} < Z_{set}$$

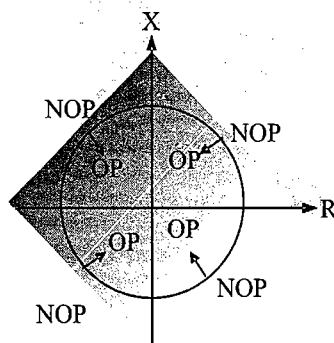


Fig. 5.11: Characteristics of impedance relay on RX diagram

Conclusions

- If the impedance seen by the relay is less than the preset value of impedance, the relay operates.
- The operation of relay depends on both resistance and reactance seen by the relay

- The impedance relay don't have inherent directional properties.
- It is moderately effected due to Arc resistance
- Occupies moderates space on RX diagram.
- Moderately effected due to power surges
- Moderately effected due to load encroachment

8.7.2 Reactance Relay

From Universal torque equation

$$T = K_1 I^2 + K_2 V^2 + K_3 VI \cos(\theta - \tau) + K$$

For Reactance Relay

$$K_1 = +ve, K_2 = 0, K_3 = -ve, K = 0$$

$$T = KI^2 - K_3 VI \cos(\theta - \tau)$$

- Current unit is providing operating torque
- Directional unit is providing restraining torque
- This relay is also called directional restrained over current relay.

For Reactance Relay maximum torque angle is 90°

$$\text{So Net torque, } T = K_1 I^2 - K_3 VI \sin \theta \dots\dots (vi)$$

From equation (vi), the reactance relay also called Reactive power restrained over current relay.

Under threshold

$$T = 0, T_{op} = T_{res}$$

$$K_1 I^2 - K_3 VI \sin \theta = 0$$

$$K_1 I^2 = K_3 VI \sin \theta$$

$$Z \sin \theta = \frac{K_1}{K_3}$$

$$X = \frac{K_1}{K_3}$$

$$X_{set} = \frac{K_1}{K_3}$$

When relay operating

$$T_{op} > T_{res}$$

$$K_1 I^2 - K_3 VI \sin \theta > 0$$

$$Z \sin \theta < \frac{K_1}{K_3}$$

$$X < \frac{K_1}{K_3}$$

$$X < X_{set}$$

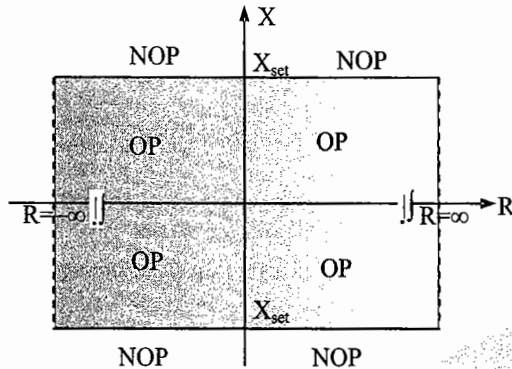


Fig. 8.12: Characteristics of Reactance relay on RX diagram

Conclusions

- If the reactance seen by the relay is less than the preset value of reactance, the relay operates
- The operation of relay depends only on reactance seen by the relay
- The reactance relay don't have inherent directional properties.
- It is not effected due to Arc resistance
- Occupies more space on RX diagram.
- Highly effected due to power surges
- Highly effected due to load encroachment

8.7.3 Mho Relay:

From Universal torque equation

$$T = K_1 I^2 + K_2 V^2 + K_3 VI \cos(\theta - \tau) + K$$

For Mho Relay

$$K_1 = 0, K_2 = -ve, K_3 = +ve, K = 0$$

$$T = K_3 VI \cos(\theta - \tau) - K_2 V^2$$

Direction unit is providing operating torque

- Voltage unit is providing restraining torque
- This relay is also called voltage restrained directional relay.

When Relay is operating

$$T > 0$$

$$K_3 VI \cos(\theta - \tau) - K_2 V^2 > 0$$

$$\frac{V}{I} < \frac{K_3}{K_2} \cos(\theta - \tau)$$

$$Z_{seen} < Z_{set}$$

$$Z_{set} = \frac{K_3}{K_2} \cos(\theta - \tau)$$

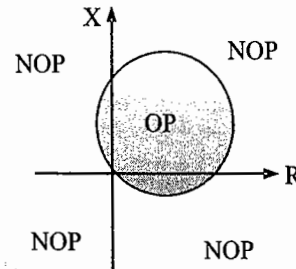


Fig. 8.13: Characteristics of Mho Relay on RX diagram

Conclusions

- If the impedance seen by the relay is less than the preset value of impedance, the relay operates
- The operation of relay depends on both resistance and reactance seen by the relay
- The Mho relay having inherent directional future.
- It is highly effected due to Arc resistance
- Occupies least space on RX diagram.
- Less effected due to power surges
- Less effected due to load encroachment

8.7.4 Application of distance Relays

Short transmission line

- The impedance measurement by the relay is effected by arc resistance because arc resistance is comparable with transmission line impedance
- Reactance relay is suitable for the protection of short transmission line because it's operation is independent on arc resistance
- Power swings on the short transmission line is very less and less severe.

Long transmission line

- Power swings are more common in long transmission line because of load variations

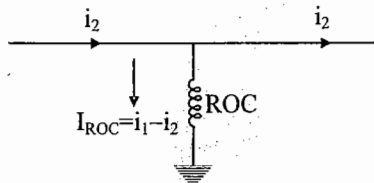
- The relay which is selected for long transmission line should be less effected due to power swings.
∴ Mho relay is preferred.
- Impedance of long line is very high so effect of arc resistance is very less on impedance measurement.

Medium transmission line:

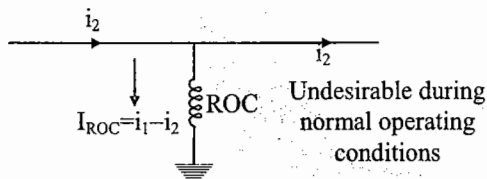
- Arc resistance is comparable with line impedance
- Power swing occurrence is appreciable
- Impedance relay is suitable.

8.8 Current differential relay

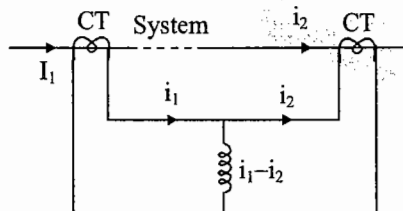
(i)



(ii)



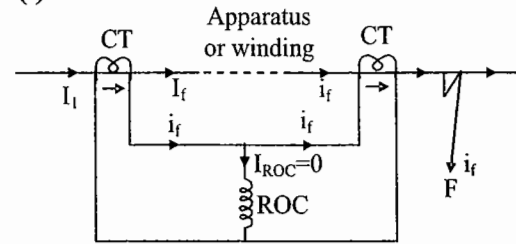
(iii)



Conditions:

1. C.T secondaries should be connected in additive polarity. Such that there is some current circulation in the C.T secondaries
2. ROC should be located exactly at mid point on pilot wire or at equal potential point.
3. Both the CT should have identical magnetization characteristics.
4. Ratio error and phase angle error should be same for both CT's

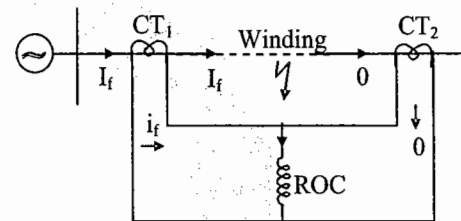
Case (i):



- Apparatus fed from only one side

Input and output currents of apparatus are same
 $I_1 = I_2 = I_R$, $I_{ROC} = 0$.

Case (ii):



Fault occurs in between 2 CT's

$I_{ROC} = I_f$ (relay operates for internal faults)

- As relay is operating only for internal fault, this protection is called unit protection.
- This relay can also be called as circulating current differential relay.
- Differential relay will have highest selectivity property compared to over current and distance relays.

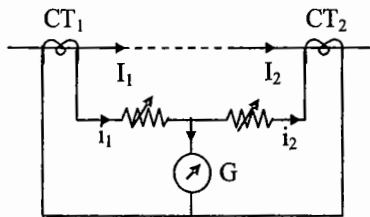
Applications:

1. stator winding of alternator
2. Transformer
3. Transmission line protection
4. Bus bar protection

Causes for mal-operation of differential relay:

1. Mislocation of ROC from equipotential points
So $I_{ROC} \neq 0$, under normal conditions relay operates.

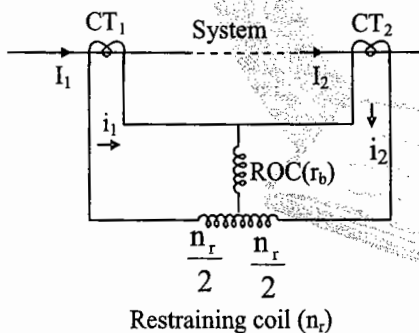
- By placing balancing resistors on the pilot wire the ROC current will be maintained at zero under normal conditions.



- Mismatch CT characteristics (BH curves)
- Difference in ratio and phase angle errors.

To avoid mal operation:

- The mal-operation of the relay is depends on system conditions.
- To restrict the relay operation, a restraining force has to be produced which should depends on system conditions.
- As operating torque depends on i_1 and i_2 the restraining torque should depends on i_1 and i_2 .
- Biased differential relay (or) %slope differential relay:**



$$I_{ROC} = i_1 - i_2$$

Torque produced by an electro magnetic coil $\propto$ (MMF produced by coil)²

$$\text{Operating torque, } T_{op} \propto [n_0(i_1 - i_2)]^2$$

$$T_{op} = K_0[n_0(i_1 - i_2)]^2$$

$$\text{Restraining torque } T_r \propto \left[\frac{n_r}{2} i_1 + \frac{n_r}{2} i_2 \right]^2$$

$$T_r \propto \left[n_r \left(\frac{i_1 + i_2}{2} \right) \right]^2$$

$$T_r = K_r \left[n_r \left(\frac{i_1 + i_2}{2} \right) \right]^2$$

$$\text{Resultant torque } T = T_{op} - T_{res} - T_s$$

T_s is inbuilt spring torque produced by over current relay element

$$T = K_0 n_0 [(i_1 - i_2)]^2 - K_r \left[n_r \left(\frac{i_1 + i_2}{2} \right) \right]^2 - K_s$$

As K_s is very very less as compared

$$K_r \left[n_r \left(\frac{i_1 + i_2}{2} \right) \right]^2$$

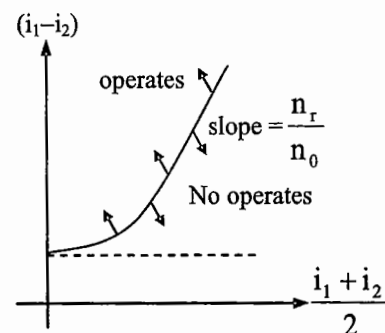
$$T = K_0 [n_0(i_1 - i_2)]^2 - K_r \left[n_r \left(\frac{i_1 + i_2}{2} \right) \right]^2$$

- If $T_{op} > T_{res}$, then relay operates,
- If $T_{op} < T_{res}$, then relay will not operate
- If $T_{op} = T_{res}$, then relay is at verge of operation.

At verge of operation, $F_{res} = 0$

$$\frac{|i_1 - i_2|}{\left(\frac{i_1 + i_2}{2} \right)} = \frac{n_r}{n_0}$$

- If $\frac{|i_1 - i_2|}{\left(\frac{i_1 + i_2}{2} \right)} > \frac{n_r}{n_0}$, then relay operates.





- Always slope provided by biased differential relay is the ratio between restraining coil turns to operation coil turns $\left(\frac{n_r}{n_o}\right)$
- Slope is adjusted by changing restraining coil turns.

Example 8.1:

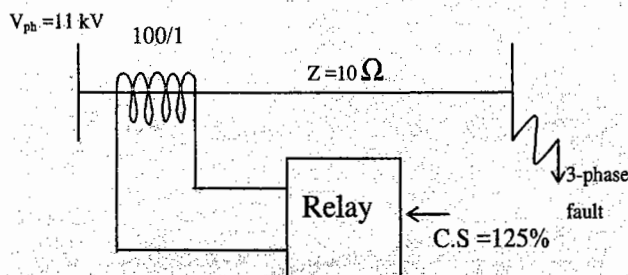
A 10 A relay is connected to the supply circuit through 1000/10 A C.T and it is set at 50 % if the fault current of the system is 5000 A, The P.S.M = ?

Sol: C.S (A) = $0.5 \times 10 = 5$ A, $I_f = 5000$ A

$$\text{P.S.M} = \frac{I_f}{\text{C.T Ratio} \times \text{C.S (A)}} = \frac{5000}{\frac{1000}{10} \times 5} = 10$$

Example 8.2:

Calculate the plug setting and plug setting multiplier for the relay as shown below.



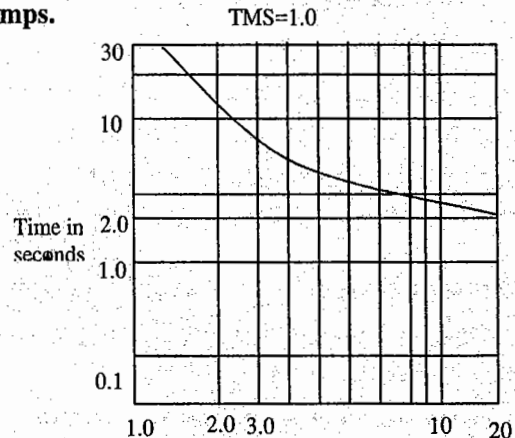
$$\begin{aligned} \text{Sol: Plug Setting} &= \frac{\text{C.S}(\%)}{100} \times I_{\text{relay}} \\ &= \frac{125}{100} \times 1 = 1.25 \text{ A} \end{aligned}$$

$$\text{Fault current } I_f = V_{ph} = \frac{11 \times 10^3}{10} = 1100 \text{ A}$$

$$\begin{aligned} \text{P.S.M} &= \frac{I_f}{\text{C.T ratio} \times \text{P.S(A)}} \\ &= \frac{1100}{\frac{100}{1} \times 1.25} = 8.8 \end{aligned}$$

Example 8.3:

Determine the time of operation of a relay of rating 5 amps, 2.2sec IDMT and having a relay setting of 125% TMS = 0.6. It is connected to a supply circuit through a C.T. 400/5 ratio. The fault current is 4000 amps.



Sol: The pick up value of the relay is 5 amps but since the relay setting is 125%, therefore, the operating current of the relay is

$$5 \times 1.25 = 6.25 \text{ amps}$$

The plug setting multiplier of the relay,

$$\begin{aligned} \text{PSM} &= \frac{\text{secondary current}}{\text{Relay Current setting}} \\ &= \frac{\text{Primary current (fault current)}}{\text{Relay current setting} \times \text{CT ratio}} \\ &= \frac{4000}{6.25 \times 80} = 8 \end{aligned}$$

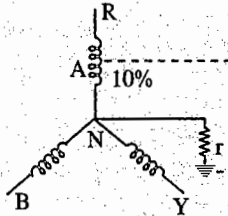
From the characteristics the operating time for PSM = 8 is 3 sec.

Since the TMS is 0.6, the actual operating time of the relay is 1.8 sec



Example 8.4:

A 6600 volt 3-phase turbo-alternator has a maximum continuous rating of 2,000 kW at 0.8 p.f. and its reactance is 12.5%. It is equipped with Merz Price circulating current protection which is set to operate at fault currents not less than 200 amperes. Find what value of the neutral earthing resistance leaves 10% of the windings unprotected?



Sol. Let the earth resistance be 'r' ohms

Full load current of the alternator

$$= \frac{2,000 \times 1,000}{0.8 \times \sqrt{3} \times 6,600} = 219 \text{ amps.}$$

Let the reactance per phase of the alternator be x ohms.

$$\therefore 12.5 = \frac{\sqrt{3} \times x \times 219}{6,600} \times 100$$

$$x = 2.19 \text{ ohms}$$

$$\therefore \text{Reactance of 10\% of the winding} = 0.219 \text{ ohm}$$

Voltage induced in winding is

$$\frac{6,600}{\sqrt{3} \times 10} = 381 \text{ volts.}$$

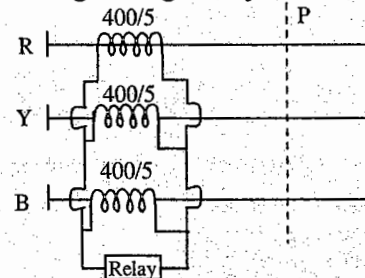
The protection operates at a current = 200 amps.

$$\therefore 200 = \frac{381}{\sqrt{r^2 + 0.219^2}}$$

Hence r = 1.89 ohms.

Example 8.5:

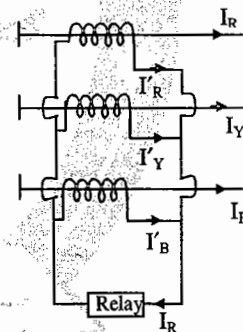
The section of balanced power system is shown below and a 3- ϕ fault occur at point 'p'. The current following through relay is



Sol: System is balance so $|I_R| = |I_Y| = |I_B|$ and phase angle between the currents is 120°

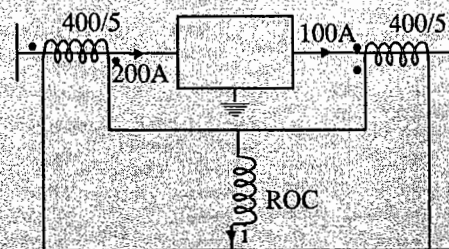
Secondary current of three C.T's are same (balanced currents)

$$\text{So } I_{\text{Relay}} = I_R^1 + I_Y^1 + I_B^1 = 0$$



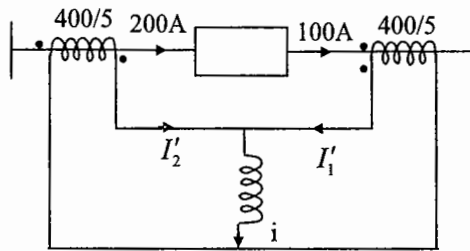
Example 8.6:

The connection of relay operating coil from two C.T's is shown below the current flowing through relay operating coil is





Sol:



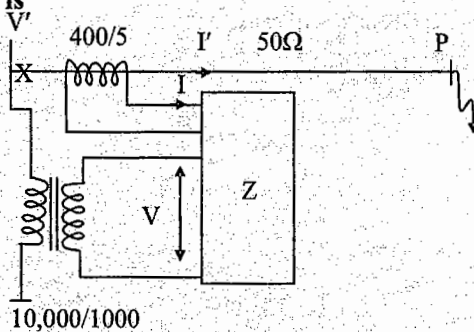
$$I_1' = 200 \times \frac{5}{400} = 2.5A$$

$$I_2' = 100 \times \frac{5}{400} = 12.5A$$

$$i = I_1' + I_2' = 3.75A$$

Example 8.7:

A distance relay connection is shown below and there is a fault at point 'p'. The impedance seen by the relay is



Sol: $Z_{seen} = \frac{V}{I}$

$$\frac{V^1}{I} = \frac{10,000}{1000} = 10 \dots (i)$$

$$\frac{I^1}{I} = \frac{400}{5} = 80 \dots (ii)$$

$$\frac{V^1}{I_1} = 50\Omega \dots (iii)$$

$$\text{Eqn. (iii)} \Rightarrow \frac{10V}{80I} = 50$$

$$\frac{V}{I} = \frac{80 \times 50}{10} = 400\Omega$$

$$Z_{seen} = 400\Omega$$